

CIRCUIT FAULT ANALYSIS REPORT

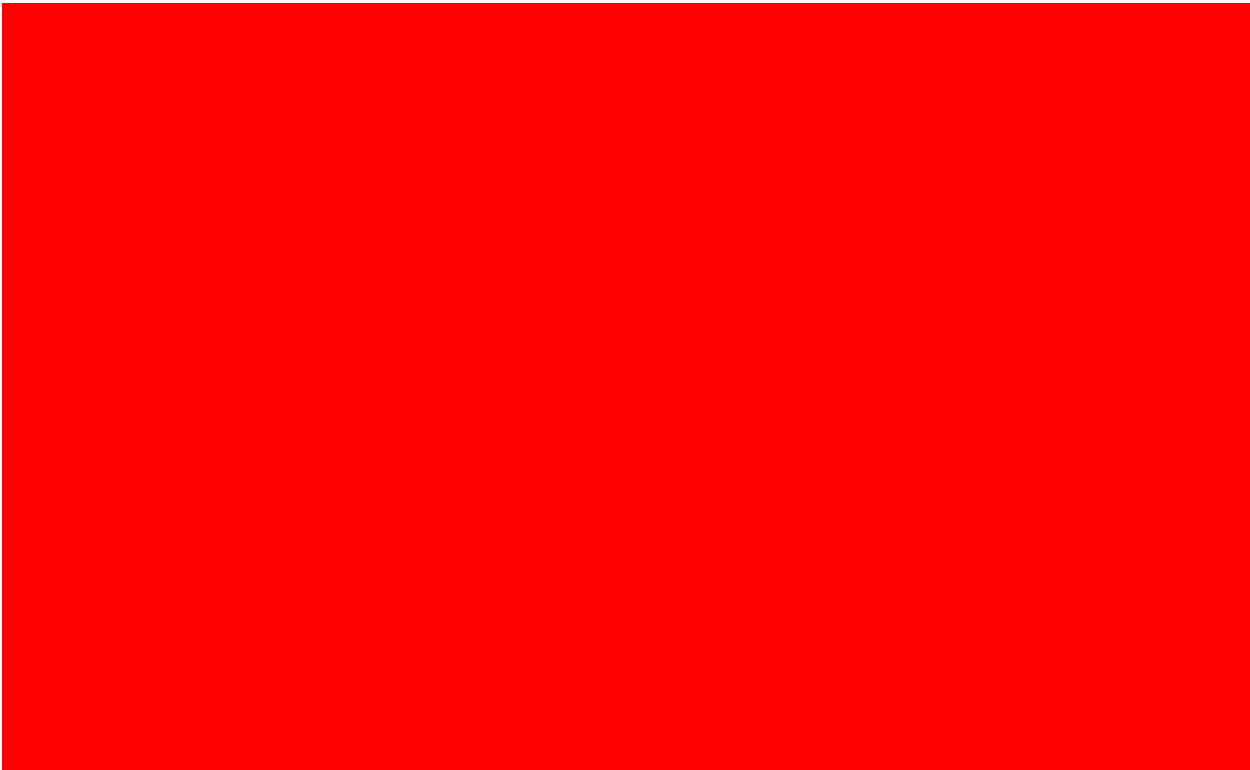
STATUS: **FAULT(S) DETECTED**

Circuit ID:	test_voltage_divider_with_image
Topology:	Voltage Divider / Series
Analysis Date:	2026-08-04 14:41:23
Generation Time:	583.90 ms

1. EXECUTIVE SUMMARY

Detected 1 fault(s) in circuit with 3 components. Primary fault type: **partial_short** (confidence: 96.0%).

2. CIRCUIT DIAGRAM



3. COMPONENT SNAPSHOT

ID	Type	Nominal	Actual	Deviation	Nodes
----	------	---------	--------	-----------	-------

V1	Dc Source	N/A	10.00 V	—	n1 ↔ 0
R1	Resistor	1000.00 Ω	1000.00 Ω	—	n1 ↔ n2
R2	Resistor	1000.00 Ω	300.00 Ω	+70.0%	n2 ↔ 0

4. SIMULATION RESULTS

4.1 Node Voltages

Node	Voltage
0	0.00 V
n1	10.000 V
n2	2.308 V

4.2 Branch Currents

Component	Current
R1	7.692×10^{-3} A
R2	7.692×10^{-3} A
V1	-7.692×10^{-3} A

5. SYMBOLIC CIRCUIT ANALYSIS

5.1 Kirchhoff's Current Law (KCL) Equations

n1: KCL at n1: $I_{V1} + I_{R1} = (V_{n1} - V_{n2})/1000.0 = 0$
n2: KCL at n2: $I_{R1} = (V_{n2} - V_{n1})/1000.0 + I_{R2} = (V_{n2} - V_0)/300.0 = 0$

5.2 Symbolic Solution

Node voltages (symbolic):
 $V_0 = 0$
 $V_{n1} = -1300.0 \cdot I_{V1}$
 $V_{n2} = -300.0 \cdot I_{V1}$

5.3 Verification

Cross-check status: FAIL

- Voltage mismatch at n1: ngspice=10.000000V, symbolic=0.000000V (diff=10.000000V)
- Voltage mismatch at n2: ngspice=2.307692V, symbolic=0.000000V (diff=2.307692V)
- Current mismatch in R1: ngspice=0.007692A, symbolic=0.000000A (diff=0.007692A)
- Current mismatch in R2: ngspice=0.007692A, symbolic=0.000000A (diff=0.007692A)

6. FAULT ANALYSIS

6.1 Fault in R2

Fault Type:	Partial Short
Severity:	CRITICAL
Confidence:	96.0%
Deviation:	+70.0%
Nominal Value:	1000.00 Ω
Actual Value:	300.00 Ω

Explanation:

A partial short fault occurs when R2 develops an unintended low-resistance path alongside its normal one, causing more current to flow than expected without fully bypassing the component. This can stem from insulation breakdown, moisture ingress, or a manufacturing defect. The actual value of 300.00 Ω compared to the nominal 1000.00 Ω (+70.0% deviation) is consistent with this partial degradation.

7. RECOMMENDATIONS

1. partial_short (Priority: 1)

- Check for solder bridges or conductive debris between component leads
- Verify resistor value with a multimeter (should match nominal $\pm 5\%$)
- Inspect PCB traces for unintended connections
- Replace component if damaged or incorrect value installed

APPENDIX

A. SPICE Netlist

```
* Circuit Fault Detector - Auto-generated Netlist
* Generated: 2026-08-04 14:41:20
* Analysis: DC Operating Point
V1 n1 0 DC 10.0
R1 n1 n2 1000.0
R2 n2 0 300.0
```

```
* Control section
.control
  op
  print v(n1)
  print v(n2)
  print v(0)
  print i(V1)
  print @R1[i]
  print @R2[i]
.endc
.end
```

B. ML Model Output

Predicted Fault Type: partial_short
Confidence: 96.00%

All Probabilities:

partial_short: 96.00%
partial_open: 47.50%
wrong_component_type: 0.00%